Brato Chakrabarti

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International Center for Theoretical Sciences
Tata Institute of Fundamental Research
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RESEARCH INTERESTS

Fluid dynamics	Fluid-structure interaction	Biophysics
Soft-matter physics	Scientific computing	Slender structures
Active matter	Biological transport	Mathematical modeling

EMPLOYMENT

Assistant Professor (*Reader-F*) — March 2024-Present
International Center for Theoretical Sciences
Tata Institute of Fundamental Research
Bengaluru, India

Flatiron Research Fellow — March 2020-January 2024
Center for Computational Biology (CCB)
Flatiron Institute, Simons Foundation
Mentor: Professor Michael J. Shelley

EDUCATION

Doctor of Philosophy, Applied Mechanics — Fall 2015-Fall 2019
Department of Mechanical and Aerospace Engineering (MAE)
University of California, San Diego
Thesis: Problems on Viscous Dynamics of Passive and Active Microfilaments
Advisor: Professor David Saintillan

Master of Science, Engineering Mechanics — Fall 2013-Spring 2015
Biomedical Engineering and Mechanics
Virginia Tech
Thesis: Catenaries in Viscous Fluid
Advisor: Professor James Hanna

Bachelor of Engineering, Mechanical Engineering — 2009-2013
Jadavpur University, India

PUBLICATIONS (* denotes equal contribution)

1. Devadevan M M and **Brato Chakrabarti** "Active Sculpting: morphodynamics of internally stirred active vesicles", arXiv:2610.19631 (2026).
2. Seema Chahal, Naveen Kumar D, and **Brato Chakrabarti** "Emergent odd response in active chiral films", arXiv:2607.17717 (2026).
3. Naveen Kumar D, Michael J. Shelley, and **Brato Chakrabarti** "Growing, Buckling, and Swirling: motility from polymerization", arXiv:2607.10446 (2026).
4. Francesco Bonacci*, **Brato Chakrabarti***, Olivia du Roure, Anke Lindner, and David Saintillan "Reversibility, Chaos, and Attractors in Periodically Sheared Elastic Filaments", *Physical Review X*, **16**, 031067 (2026).
5. Jasmin Imran Alsous*, **Brato Chakrabarti***, Bryce Palmer, and Michael J. Shelley "The physical consequences of sperm gigantism", *Nature Physics*, **22**, 1306–1317 (2026).

6. Dipanjan Ghosh, **Brato Chakrabarti**, and Xiang Cheng "Density-controlled reversal of collective migration in bacterial aerotaxis", *Physical Review Research*, **8**, 033059 (2026).
7. Seema Chahal and **Brato Chakrabarti** "Hydrodynamic instabilities in driven chiral suspensions", *Physical Review E Letters*, **113** (2), L023101 (2026).
8. Olenka Jain, **Brato Chakrabarti**, Reza Farhadifar, Elizabeth R. Gavis, Michael J. Shelley, and Stanislav Y. Shvartsman "Geometric Effects in Large Scale Intracellular Flows", *PRX Life*, **3** (2), 023007 (2025).
9. **Brato Chakrabarti**, Manas Rachh, SY Shvartsman, and Michael J. Shelley "Cytoplasmic stirring by active carpets", *Proceedings of the National Academy of Sciences*, **121**, e2405114121 (2024).
10. Chenji Li, **Brato Chakrabarti**, Pedro Castilla, Achal Mahajan, and David Saintillan, "Chemomechanical model of sperm locomotion reveals two modes of swimming", *Physical Review Fluids*, **8** 113102 (2023).
11. **Brato Chakrabarti**, Michael J. Shelley, and Sebastian Fürthauer "Collective Motion and Pattern Formation in Phase-Synchronizing Active Fluids", *Physical Review Letters* **130**, 128202 (2023).
12. Francesco Bonacci, **Brato Chakrabarti**, David Saintillan, Olivia du Roure, and Anke Lindner "Dynamics of flexible filaments in oscillatory shear flows", *Journal of Fluid Mechanics*, **955** A35 (2023).
13. A. C. Quillen, A. Peshkov, **Brato Chakrabarti**, Nathan Skerrett, Sonia McGaffigan, and Rebeca Zapiach "Fluid circulation driven by collectively organized metachronal waves in swimming *T. aceti* nematodes", *Physical Review E*, **106** 064401 (2022).
14. **Brato Chakrabarti**, Sebastian Fürthauer and Michael J. Shelley, "A multiscale biophysical model gives quantized metachronal waves in a lattice of beating cilia", *Proceedings of the National Academy of Sciences of the USA* **119**, e2113539119 (2022).
15. **Brato Chakrabarti**, Yanan Liu, Olivia du Roure, Anke Lindner, and David Saintillan, "Signatures of elastoviscous buckling in the dilute rheology of stiff polymers", *Journal of Fluid Mechanics*, **919** A12 (2021).
16. **Brato Chakrabarti**, and David Saintillan, "Shear-induced dispersion in peristaltic flow", *Physics of Fluids*, **32** 113102 (2020). **Invited:** "Contributions from Early Career Researchers 2020" and selected as a **featured** article.
17. **Brato Chakrabarti**, Charles Gaillard, and David Saintillan, "Trapping, gliding, vaulting: Transport of semiflexible polymers in periodic post arrays", *Soft Matter*, **16** 5534–5544 (2020).
18. **Brato Chakrabarti**, Yanan Liu, John LaGrone, Ricardo Cortez, Lisa Fauci, Olivia du Roure, David Saintillan, and Anke Lindner, "Flexible filaments buckle into helicoidal shapes in strong compressional flows", *Nature Physics*, **16** 689–694 (2020).
19. **Brato Chakrabarti** and David Saintillan, "Hydrodynamic synchronization of spontaneously beating filaments", *Physical Review Letters*, **123** 208101 (2019).
20. **Brato Chakrabarti** and David Saintillan, "Spontaneous oscillations, beating patterns, and hydrodynamics of active microfilaments", *Physical Review Fluids*, **4** 043102 (2019).
21. Roberto Alonso-Matilla, **Brato Chakrabarti** and David Saintillan, "Transport and dispersion of active particles in periodic porous media", *Physical Review Fluids*, **4** 043101 (2019).
22. Yanan Liu*, **Brato Chakrabarti***, David Saintillan, Anke Lindner and Olivia du Roure, "Morphological transitions of elastic filaments in shear flow", *Proceedings of the National Academy of Sciences of the USA*, **115** 9438–9443 (2018).
23. **Brato Chakrabarti** and James Hanna "Catenaries in Viscous Fluid", *Journal of Fluids and Structures*, **66** 490–516 (2016).

CONFERENCE ARTICLES AND PRESENTATIONS (presenter underlined)

1. Seema Chahal and Brato Chakrabarti, 'Hydrodynamic instabilities in driven chiral suspensions', at the 78th Annual Meeting of the APS Division of Fluid Dynamics, November 2025, Houston, USA (**Selected for interactive session**).
2. Seema Chahal and Brato Chakrabarti, 'Hydrodynamic instabilities in driven chiral suspensions', 10th Indian Statistical Physics Community Meeting, ICTS-TIFR, 2024.
3. Francesco Bonacci, Brato Chakrabarti, Olivia du Roure, Anke Lindner, and David Saintillan, *Reversible to chaotic transitions in the dynamics of fluctuating elastic filaments in oscillatory shear flow*, at the 77th Annual Meeting of the APS Division of Fluid Dynamics, November 2024, Salt Lake City, USA.

4. Brato Chakrabarti, 'Cytoplasmic streaming by active carpets', 9th Indian Statistical Physics Community Meeting, ICTS-TIFR, 2024.
5. Brato Chakrabarti, and Michael J Shelley, *A coarse-grained model for cytoplasmic streaming*, at the 76th Annual Meeting of the APS Division of Fluid Dynamics, November 2023, Washington DC, USA.
6. Brato Chakrabarti, and Michael J Shelley, *A coarse-grained model for cytoplasmic streaming*, at the APS March Meeting, 2023, Las Vegas, USA.
7. Brato Chakrabarti, Sebastian Fürthauer and Michael J Shelley, *Self-organized flows in phase-synchronizing active fluids*, at the APS March Meeting, 2022, Chicago, USA.
8. Brato Chakrabarti, Sebastian Fürthauer and Michael J Shelley, *A multiscale biophysical model gives quantized metachronal waves in a lattice of cilia*, at the APS March Meeting, 2022, Chicago, USA.
9. Francesco Bonacci, Brato Chakrabarti, Olivia du Roure, Anke Lindner, and David Saintillan, *Dynamics of semiflexible filaments in oscillatory shear flows*, at the Annual European Rheology Conference, Sevilla, 2022.
10. Brato Chakrabarti, Sebastian Fürthauer and Michael J Shelley, *A multiscale biophysical model gives quantized metachronal waves in a lattice of cilia*, at the 74th Annual Meeting of the APS Division of Fluid Dynamics, November 2021, Phoenix, USA.
11. David Saintillan, Chenji Li, Brato Chakrabarti, Pedro Castilla, and Achal Mahajan *An integrated chemomechanical model of sperm locomotion reveals two fundamental swimming modes*, at the 74th Annual Meeting of the APS Division of Fluid Dynamics, November 2021, Phoenix, USA.
12. David Saintillan, Yanan Liu, John Lagrone, Ricardo Cortez, Lisa Fauci, Olivia du Roure, Anke Lindner, and Brato Chakrabarti *Viscous dynamics of elastic filaments: from buckling instabilities to rheology*, at the APS March Meeting, 2021 (online).
13. Brato Chakrabarti, Yanan Liu, Olivia du Roure, Anke Lindner, and David Saintillan, *Signatures of elastoviscous buckling in the dilute rheology of stiff polymers*, at the 73rd Annual Meeting of the APS Division of Fluid Dynamics, November 2020 (online).
14. David Saintillan, and Brato Chakrabarti, *Hydrodynamic synchronization of spontaneously beating filaments*, at the 72nd Annual Meeting of the APS Division of Fluid Dynamics, November 2019, Seattle, USA.
15. Brato Chakrabarti, Yanan Liu, John Lagrone, Ricardo Cortez, Lisa Fauci, Olivia du Roure, David Saintillan, and Anke Lindner *Helical buckling of flexible filaments in viscous flow*, at the 72nd Annual Meeting of the APS Division of Fluid Dynamics, November 2019, Seattle, USA.
16. Anke Lindner, Brato Chakrabarti, Yanan Liu, Olivia du Roure and David Saintillan, *The dynamics of flexible Brownian fibers in viscous flows* at The Annual European Rheology Conference, Slovenia, April 8-11, 2019.
17. Brato Chakrabarti and David Saintillan, *Spontaneous oscillations and hydrodynamics of active micro-filament* at the 71st Annual Meeting of the APS Division of Fluid Dynamics, November 2018, Atlanta, USA.
18. Roberto Alonso-Matilla, Brato Chakrabarti and David Saintillan, *Asymptotic transport and dispersion of active particles in periodic porous media* at the 71st Annual Meeting of the APS Division of Fluid Dynamics, November 2018, Atlanta, USA.
19. Brato Chakrabarti, Yanan Liu, David Saintillan, Anke Lindner and Olivia du Roure, *The dynamics of flexible and Brownian filaments in viscous flows* at the 71st Annual Meeting of the APS Division of Fluid Dynamics, November 2018, Atlanta, USA.
20. Brato Chakrabarti, Yanan Liu, David Saintillan, Anke Lindner and Olivia du Roure, *Buckling and migration of semi-flexible filaments* at the 70th Annual Meeting of the APS Division of Fluid Dynamics, November 2017, Denver, USA.
21. David Saintillan and Brato Chakrabarti, *Shear dispersion in peristaltic pumping* at the 70th Annual Meeting of the APS Division of Fluid Dynamics, November 2017, Denver, USA.
22. James Hanna and Brato Chakrabarti, *Catenaries in viscous fluid*. 24th ICTAM, Montreal, August 2016.
23. Brato Chakrabarti and David Saintillan. *Drift, Mixing and Diffusivity in Stokes Flow*. Presented at the Southern California (SoCal) Fluids X, April 2016, UC Irvine, California, USA.
24. Brato Chakrabarti and James Hanna. *Catenaries in viscous fluid*. At the 68th Annual Meeting of the APS Division of Fluid Dynamics, November 2015, Boston, USA.
25. Brato Chakrabarti and James Hanna. *Catenaries in Drag*. Presented at the 67th Annual Meeting of the APS Division of Fluid Dynamics, November 2014, San Francisco, USA.

INVITED TALKS

1. 'Hydrodynamic instabilities in bacterial carpet', Indian Mechanobiology Meeting, IIT-Hyderabad, October, 2026.
2. 'Challenges in Biophysical Modeling', Mathematics of the real world: Research–Industry–Academia Workshop, IIT-Bombay, August, 2026.
3. 'The physical consequence of sperm gigantism', Mini-Symposium on AI for Biology, NCBS, Bengaluru, August, 2026.
4. 'The physical consequence of sperm gigantism', Complexity in Matter and Policy, Indian Institute of Science, IISc, Bengaluru, August, 2026.
5. 'The physical consequence of sperm gigantism', 41st Ramanujan Mathematical Society Conference, July, 2026.
6. 'The physical consequence of sperm gigantism', International Soft Matter Conference 2026, Goa, May, 2026.
7. 'The physical consequence of sperm gigantism', IntCha26, Cargese, France, April, 2026.
8. 'Odd response in chiral thin films', Department of Mechanical Engineering, IIT Kanpur, April, 2026
9. 'The physical consequence of sperm gigantism', Current Trends in Active Matter Physics 3.0, IISc, Bengaluru, March, 2026.
10. 'The physical consequence of sperm gigantism', Modelling and Tackling Complex Biological Systems 2.0, IMSc, Chennai, February, 2026.
11. 'The physical consequence of sperm gigantism', StatPhys Kolkata XIII, SNBNCBS, Kolkata, January, 2026.
12. 'The physical consequence of sperm gigantism', Mechanics of Life III, Flatiron Institute, New York, December, 2025.
13. 'Boundary-driven self-organized flows: discrete simulations and continuum theories', Department of Chemical Engineering, Indian Institute of Science, October, 2025.
14. 'Living fluids across scales', Soft Matter Young Investigator Meet (SMYIM), Shimla, June, 2025.
15. 'The physical consequence of sperm gigantism', Theory and Concepts in Biology, EMBL, Heidelberg, 2025.
16. 'Active carpet model for intracellular flows', CompFlu, Indian Institute of Technology, Hyderabad, 2024.
17. 'From one to many: two problems in fluid-structure Interactions at low Reynolds number', LadHyX, Ecole Polytechnique, France, 2024.
18. 'Problems and puzzles in intracellular flows', PMMH-ESPCI, Paris, France, 2024.
19. 'Self-organized intracellular flows: coarse-grained theory and the role of geometry', Symposium on Nonequilibrium and Active Matter Physics Meeting, Indian Institute of Science, 2024.
20. 'Self-organized intracellular flows: computational methods and coarse-grained theory', International Workshops on Advances in Computational Mechanics - IV, Kitakyushu, Japan, 2024.
21. 'Cytoplasmic streaming by active carpets', Mechanical Science Young Investigator Meeting (MSYIM), IIT Kanpur, 2024.
22. 'The waves within us: hydrodynamics of passive and active filaments', Indian Association for the Cultivation of Science, Kolkata, India, 2024.
23. 'The waves within us: hydrodynamics of passive and active filaments', University of California, Riverside, 2023.
24. 'Quantized metachronal waves in arrays of cilia', BPPB seminar (online), 2023.
25. 'Beat, sync, and wave: nonlinear dynamics of flagella and cilia', Department of Mechanical Engineering, IIT Bombay, 2023.
26. 'The waves within us: hydrodynamics of passive and active filaments', Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR), Bengaluru, 2023.
27. 'The waves within us: nonlinear dynamics of passive and active filaments', Tata Institute of Fundamental Research, Hyderabad, 2023.
28. 'A continuum theory for cytoplasmic streaming in the *Drosophila* oocyte', International Center for Theoretical Sciences (ICTS), Bengaluru, 2022.
29. 'Beat, sync, and wave: nonlinear dynamics of cilia and flagella', Colloquium, International Center for Theoretical Sciences (ICTS), Bengaluru, 2022.
30. 'Problems on nonlinear dynamics of filaments in viscous fluids', Department of Applied Mechanics, Indian Institute of Technology, Madras, 2022.
31. 'From buckling to streaming: problems on fluid-structure interaction in viscous flows', Department of Mathematics, Princeton University, Analysis of Fluids Seminar, 2022.

32. 'The waves within us: problems on dynamics of passive and active filaments', International Center for Theoretical Sciences (ICTS), Bengaluru, 2022.
33. 'Problems on viscous dynamics of passive and active filaments: from one to many', Raman Research Institute (RRI), Bengaluru, 2022.
34. 'The waves within us: from single cilium to the formation of metachronal waves', Institute of Mathematical Sciences (IMSc), Madras, 2022.
35. 'The waves within us', Simons Foundation Lecture Series, Flatiron Institute, 2022.
36. 'A multiscale biophysical model gives quantized metachronal waves in a lattice of cilia', Frontiers in Applied & Computational Mathematics, New Jersey Institute of Technology, 2022.
37. 'Hydrodynamics of Active Matter', Jadavpur University, 2021.
38. 'Metachronal waves in ciliary arrays', Brown Bag Seminar, Center for Computational Biology, Flatiron Institute, 2020.
39. 'Helical buckling of actin filaments in compressional flow', Biophysical Modeling group, Center for Computational Biology, Flatiron Institute, 2019.
40. 'Viscous dynamics of active and passive microfilaments', Department of Physics, University of California, Santa Barbara, 2019.
41. 'Spontaneous oscillations and hydrodynamic synchronization of active filaments', ESPCI, Paris, France 2019.

GRANTS

1. Prime Minister Early Career Research Grant awarded by the Anusandhan National Research Foundation, 2025.

STUDENTS MENTORED

Postdoctoral Fellows

1. Shreyank Goel, CALIBRE Postdoctoral Fellow (Fall 2026 - Ongoing)

Ph.D. Students

1. Naveen Kumar D, Physics (January 2025 - Ongoing)
2. Anjali Kundalpady, Physics (co-supervised with Prof. Rama Govindarajan) (January 2025- Ongoing)
3. Devadevan M M, Physics (January 2025 - Ongoing)
4. Seema Chahal, Physics (co-supervised with Prof. Anupam Kundu) (January 2025 - Ongoing)
5. Krishnadev V, Physics of Life (October 2025 - Ongoing)

Visiting Students

1. Saranya S, IISER Tirupati (Visiting student, Summer 2026)
2. Reet Jaiswal, IISER Mohali (S. N. Bhatt student 2025)
3. Sidarth S, IIT M (LTVSP 2024)

OUTREACH

- Introduction to hydrodynamics and life in moving fluids, Invited pedagogical lectures at the C. K Majumdar workshop, SNBNCBS, Kolkata (2026)
- Beyond Boundaries: A Day of Interdisciplinary Exploration, ICTS (2024)
- Invited talk at Physics of Life Monsoon School, NCBS, Bengaluru (2024)
- Instructor at the summer course on Dynamics of Biological Systems, ICTS-TIFR, Bengaluru (2024)

SERVICE

- Subject board representative of Biology from ICTS-TIFR.
- Coordinator of the Physics of Life graduate program at ICTS-TIFR.
- Member of the graduate cell at ICTS-TIFR.

ORGANIZATION

- Co-organizer of the session on Collective Behaviors in Biology, APS Global Summit, 2026.
- Co-organizer of the session on Computational Fluid Dynamics at the 41st Annual Conference of Ramanujan Mathematical Society, July, 2026.
- Co-organizer of the ICTS Discussion Meeting on Multiscale Flows and Self-Organization in Living Matter, March, 2027.

AWARDS AND HONORS

1. **Scientific High-Level Visiting Fellowships** from the French Institute of India, 2024.
2. **Early Career Researcher** awarded by the journal *Physics of Fluids*, 2020.
3. **Powell Fellow, UCSD** by Jacobs school of Engineering, 2015.
4. **Awarded Gold Medal** for best performance in Fluid Mechanics in Bachelor of Engineering (Mechanical Engineering, Jadavpur University), 2013.
5. **National Merit Scholarship** for performance in school leaving examination, 2009.

REVIEWER OF ARCHIVED JOURNALS

1. Journal of Fluid Mechanics
2. Physical Review Letters
3. Physical Review Fluids
4. Physical Review E
5. Soft Matter
6. Journal of Computational Physics
7. Physica D
8. New Journal of Physics
9. Journal of Mathematical Fluid Mechanics
10. Proceedings of the Royal Society A
11. Proceedings of the National Academy of Sciences
12. Nature Communications
13. Communications Physics
14. Physical Review Research

EDITORIAL WORK

- Editorial Board Member (EBM), Communications Physics, Nature

TEACHING EXPERIENCE & OUTREACH

ICTS-TIFR

- Fall 2024: Fluid dynamics and elasticity
- Spring 2025: Biological fluid dynamics and active suspensions
- Fall 2025: Advanced fluid dynamics
- Fall 2026: Physics of Living Matter

Virginia Tech

- Fall 2013: Teaching Assistant, Statics (ESM 2104)
- Spring 2014: Teaching Assistant, Dynamics (ESM 2204)
- Fall 2014: Teaching Assistant, Analytical mechanics (ESM 3214)
- Spring 2015: Teaching Assistant, Vibrations (ESM 3134)

UCSD

- Winter 2017: Teaching Assistant, Fluid mechanics (MAE 210 A)
- Fall 2018: Teaching Assistant, Introduction to mathematical physics (MAE 105).
- Spring 2019: Teaching Assistant, Hydrodynamic stability (MAE 210 C)